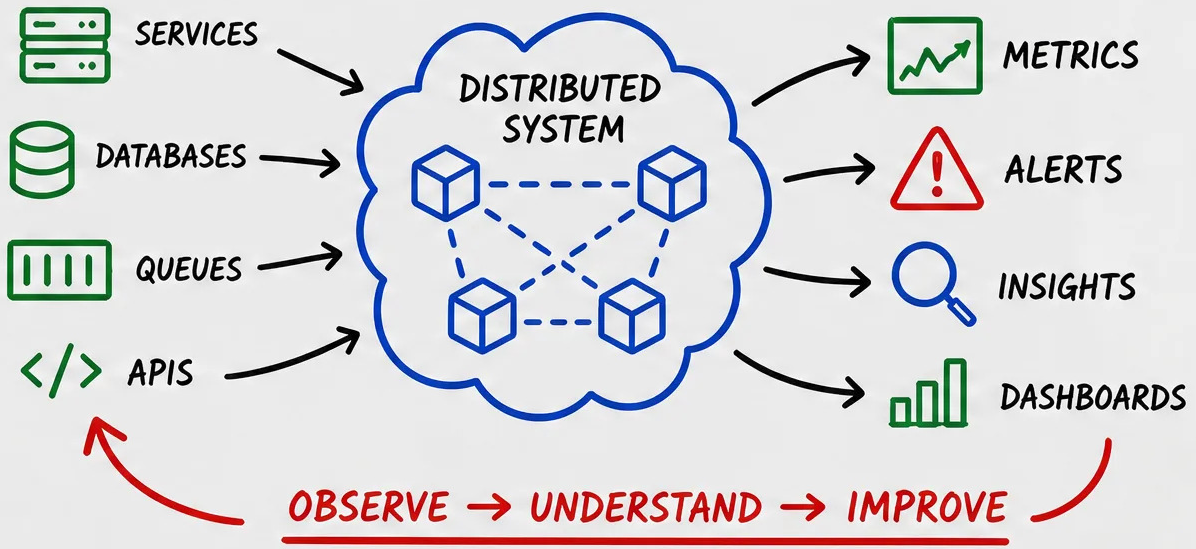
ProTech
protechtraining.com

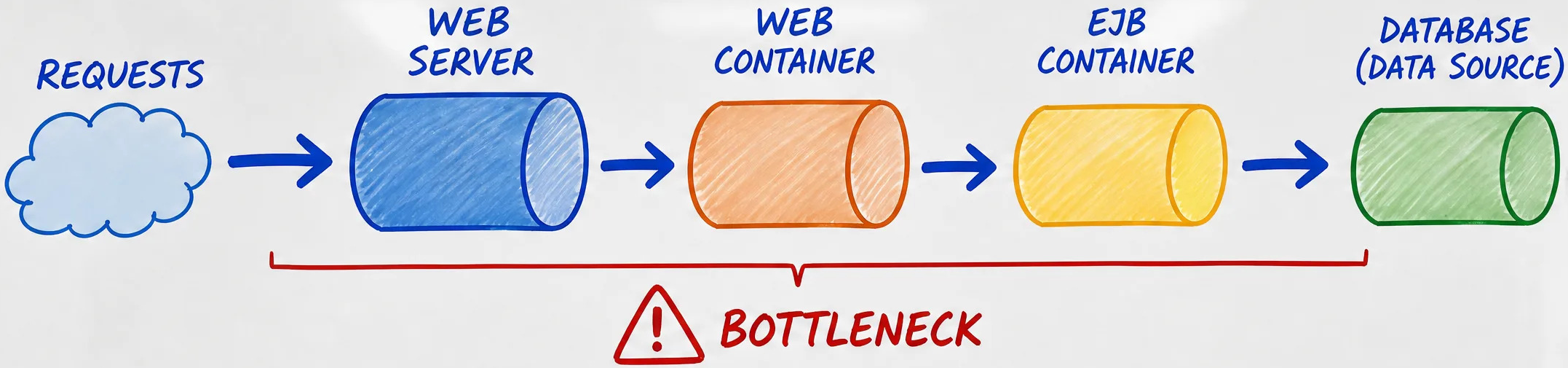
Module 16 — Monitoring Distributed Systems

Collect once, consume wisely

Metrics earn their keep by answering *questions*.



Define a Healthy Request End to End



LATENCY HAS A PATH

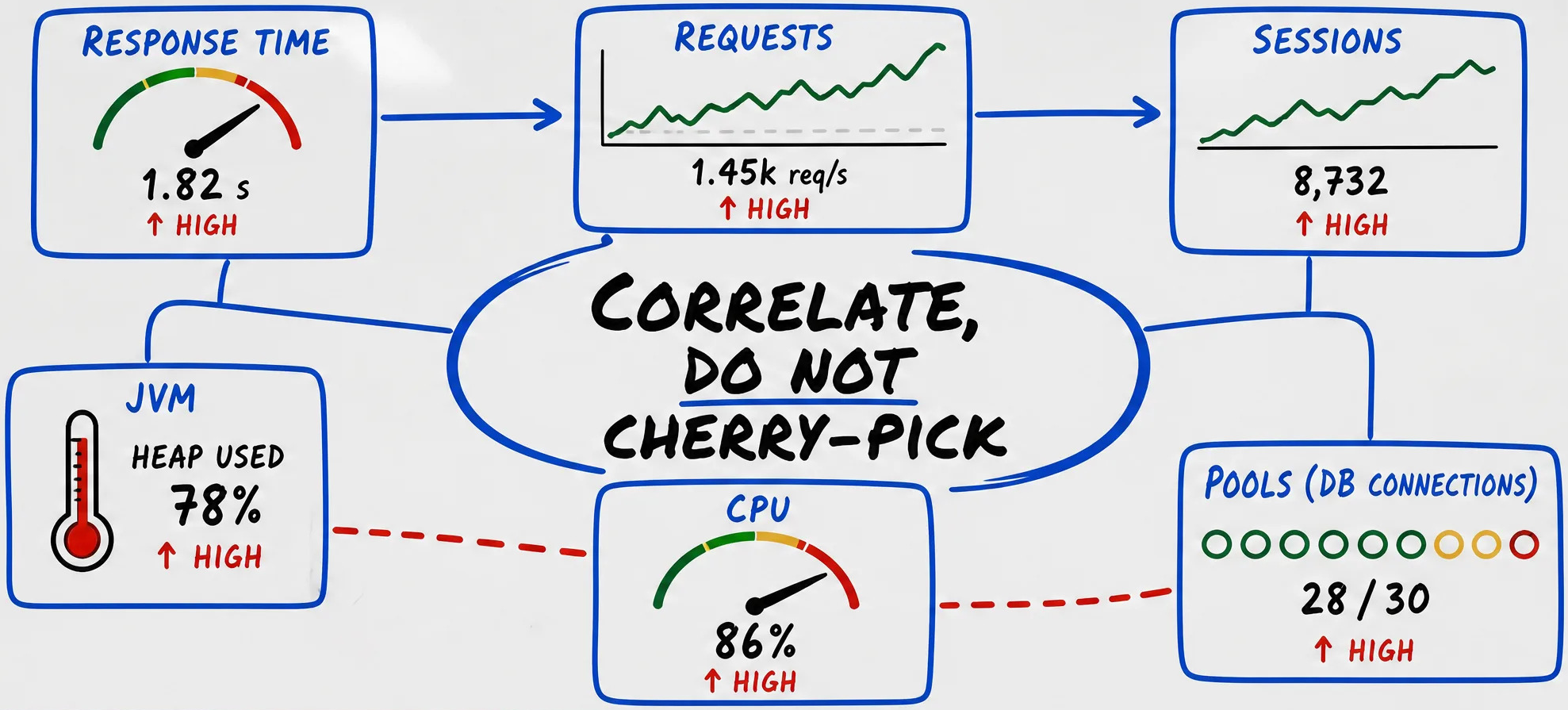
A HEALTHY SERVER CAN STILL SERVE AN UNHEALTHY REQUEST.

PMI Turns Runtime Into Evidence



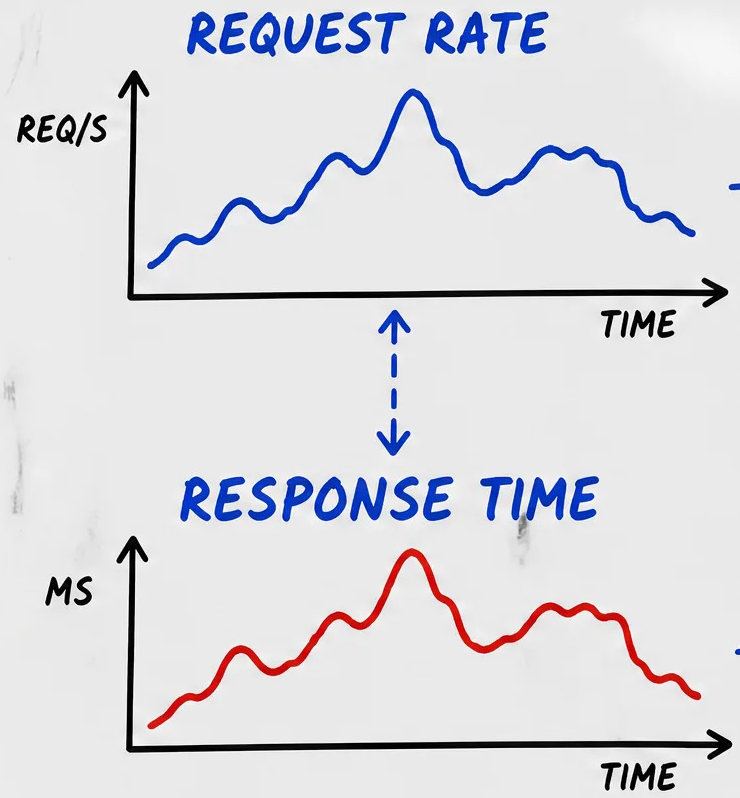
PMI is the **measurement plane**, not the **tuning decision**.

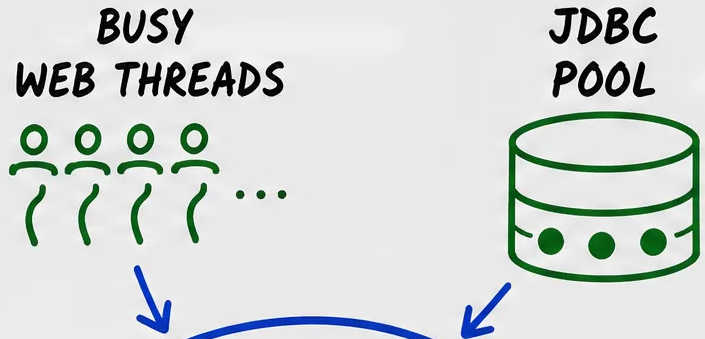
Choose Health Signals



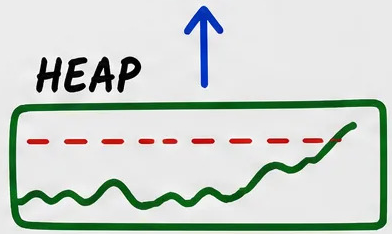
One metric notices trouble; **relationships** explain it.

CORRELATE BEFORE YOU RESIZE





SYMPTOMS SHARE CAUSES



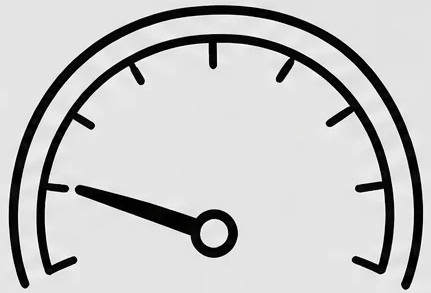
NOT THE ANSWER:

RANDOM KNOB TURN



A FULL POOL IS **EVIDENCE**, NOT AN AUTOMATIC **RESIZE ORDER.**

Start Lean With PMI



NONE

No counters enabled



BASIC

Start here.
Lowest overhead.



EXTENDED

More visibility.
More overhead.



ALL / CUSTOM

Highest overhead.
Use with caution.

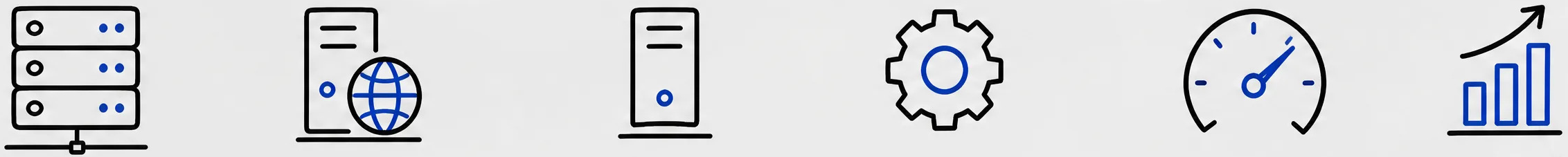
EVERY COUNTER HAS A COST



ENABLE ONLY WHAT ANSWERS THE NEXT OPERATIONAL QUESTION.

Enable, Save, Restart

Servers → Application servers → Server → Configuration → Performance → PMI







PMI configuration **does not take effect** until the server restarts.

Verify Collection First

Preflight Checklist

- Restart ✓
- Selected set ✓
- Advancing timestamp ✓
- Known request ✓
- Visible counter ✓

No evidence,

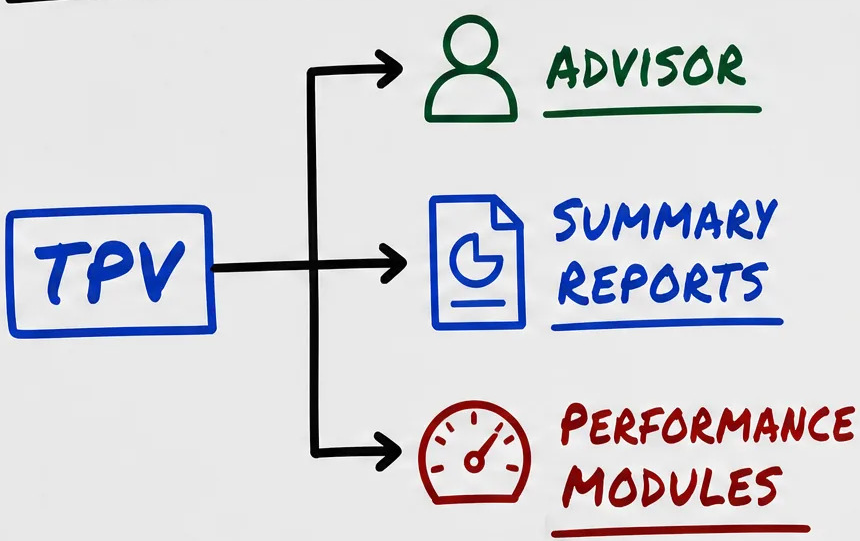
no diagnosis.



A chart without provenance is **not** operational evidence.

Observe With TPV

TPV Navigation



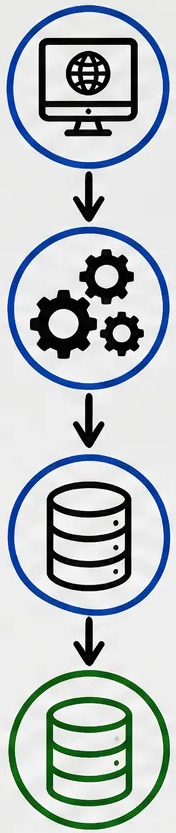
Select the Question, Then Counters





TPV makes PMI visible; it does not interpret causation for you.

Read Pools as a Pipeline



REQUEST — Incoming work arrives

THREAD — Thread from pool processes request

JDBC — JDBC connection executes SQL against DB

DATABASE — Database processes the query

Follow work downstream

Baseline (now)

May 16, 2025
10:15 AM

Application servers > **AppSrv_85_01** > **Thread pools**

Use this page to specify a thread pool for the server to use. A thread pool enables server components to reuse threads instead of creating new threads at run time. Creating new threads is typically a time and resource intensive operation.

Preferences

New... Delete

Select	Name	Description	Minimum Size	Maximum Size
You can administer the following resources:				
☐	Default		20	20
☐	ORB.thread.pool		10	50
☐	SIBFAPInboundThreadPool	Service integration bus FAP inbound channel thread pool	4	50
☐	SIBFAPThreadPool	Service integration bus FAP outbound channel thread pool	4	50
☐	SIBJMSRAThreadPool	Service Integration Bus JMS Resource Adapter thread pool	35	41
☐	TCPChannel.DCS		20	20
☐	WMQJCAResourceAdapter	WebSphere MQ Resource Adapter thread pool	10	50
☐	WebContainer		50	50
☐	server.startup	This pool is used by WebSphere during server startup.	1	3

Total 9

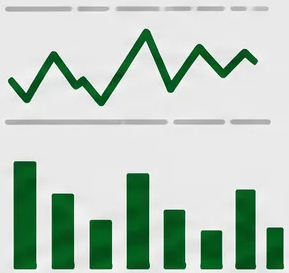
⚠ **Downstream idleness can reveal an upstream constraint.**

Treat Advice as a Hypothesis



WHITEBOARD ADVISOR

PMI DATA



Collect real signals, metrics, and context.

ADVICE NEEDS A WORKLOAD

TEST THE HYPOTHESIS



WARNING

Don't tune on idle or unrepresentative conditions.

BASELINE



Measure current behavior under representative load.

RETEST



Measure again under the same load to validate impact.

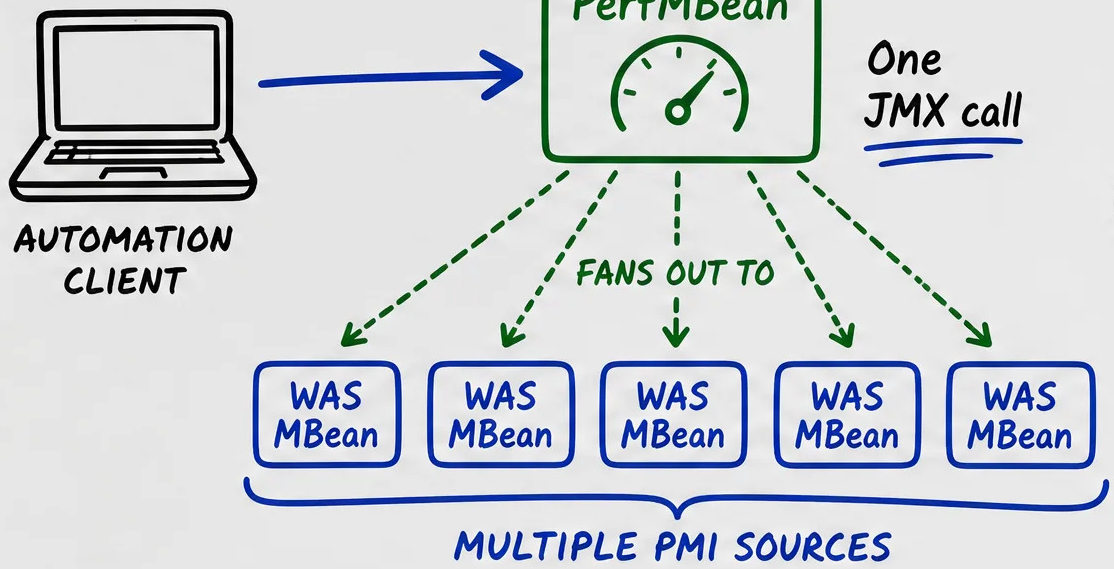


A TUNING RECOMMENDATION IS **CREDIBLE ONLY** UNDER **REPRESENTATIVE LOAD.**

AUTOMATE WITH JMX

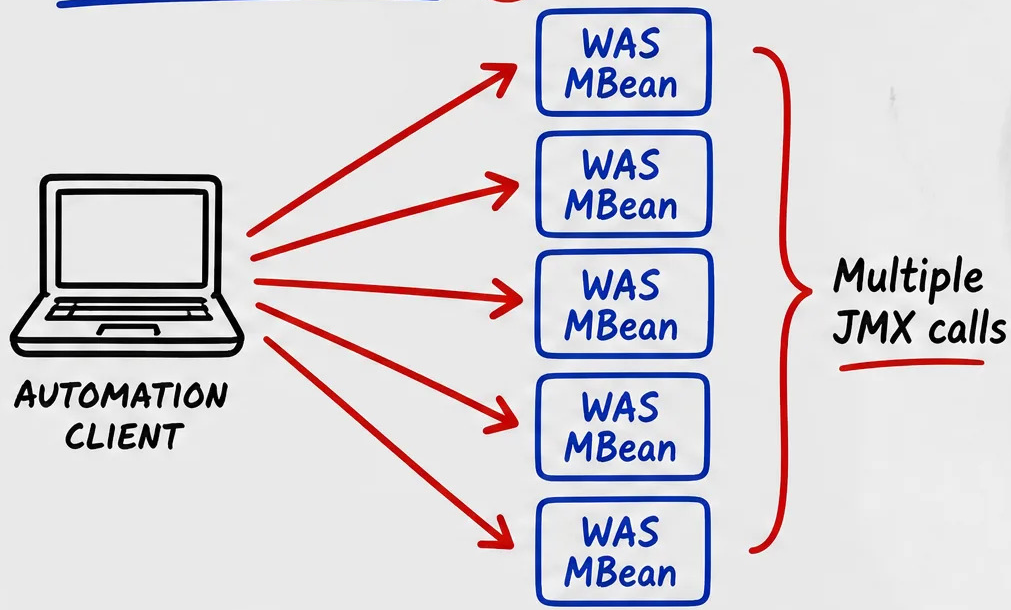
USE THE SUPPORTED MEASUREMENT API

ONE CALL ✓



VS.

MANY CALLS ✗



PERFMBEAN CAN COLLECT MULTIPLE PMI SOURCES THROUGH ONE JMX CALL.

Draw the PerfServlet Boundary

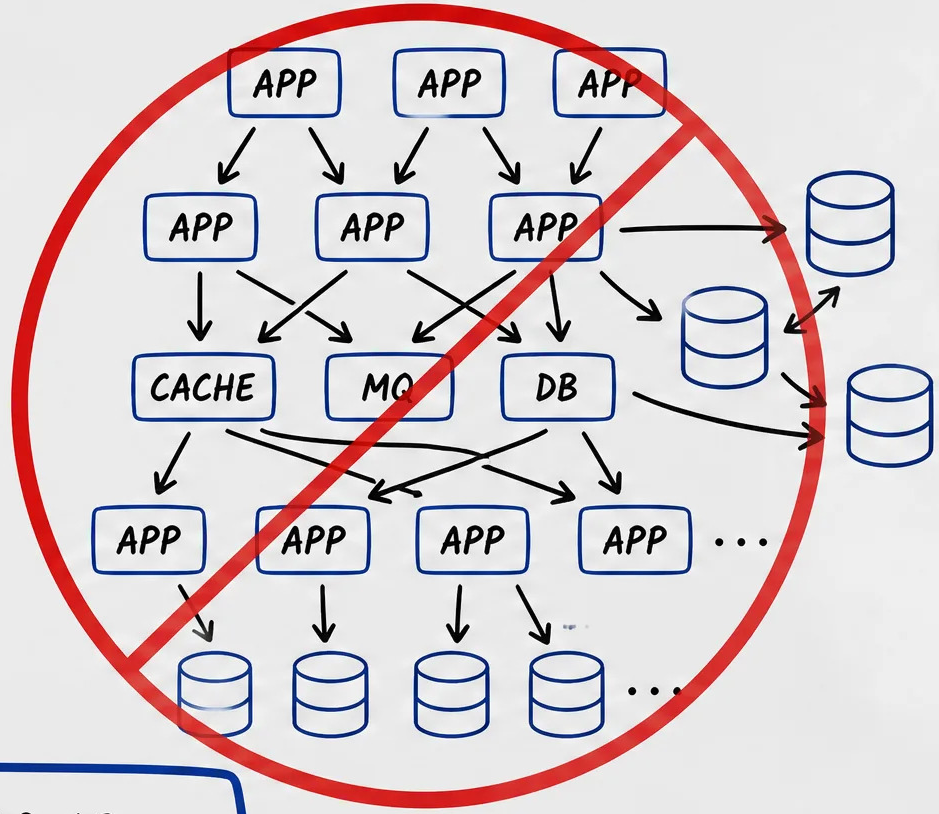
Sample-tool test bench



- Single-threaded
- One JVM
- Local scope
- Good for testing

SAMPLE IS NOT SCALE

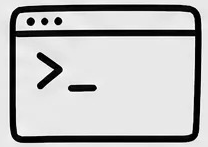
Production topology



A **SINGLE-THREADED** sample collector is **NOT** a production monitoring architecture.

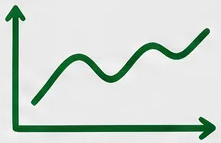
Modernize Without Erasing

Legacy	Transition	Modern
PMI Client API	Replace the mechanism,	JMX PerfMBean
SVG Graphs	Keep the evidence.	Dojo
TPV		TPV (remains)



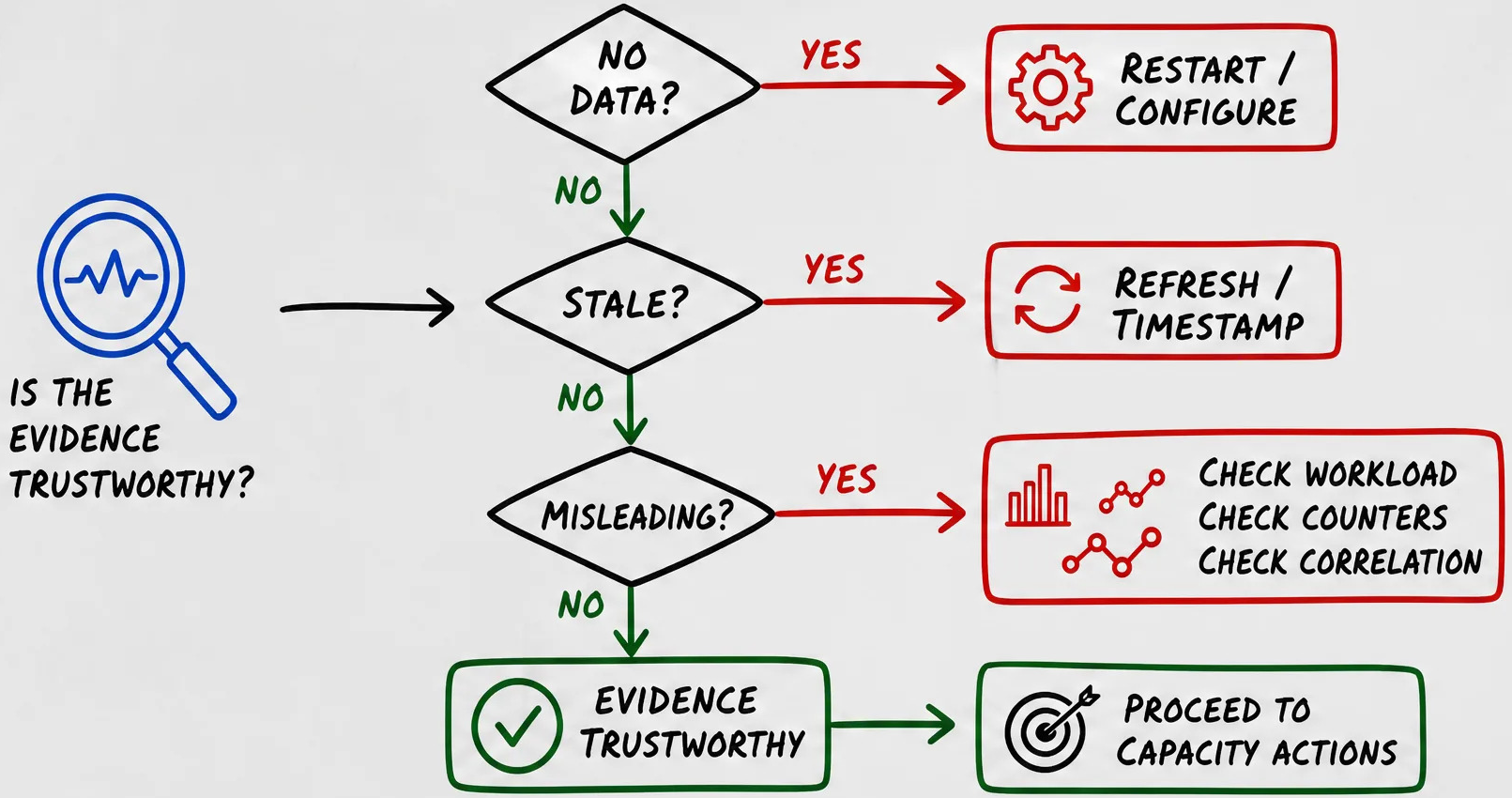






⚠ **DEPRECATE** an interface, **NOT** the need to measure.

Triage Monitoring Evidence



FIX EVIDENCE BEFORE CAPACITY

DO NOT TUNE A SERVER UNTIL YOU **TRUST ITS MEASUREMENTS.**

Lab: Prove a Monitoring Decision

Enable → Turn on monitoring

Restart → Apply change & restart

Verify → Confirm data is flowing

Load → Generate baseline

Compare → Compare to baseline

Hypothesis → Form hypothesis & predict outcome

Measure, explain, then change.

Baseline

Rollback plan

If risk increases, roll back.

The lab succeeds when **evidence** earns the **next action.**